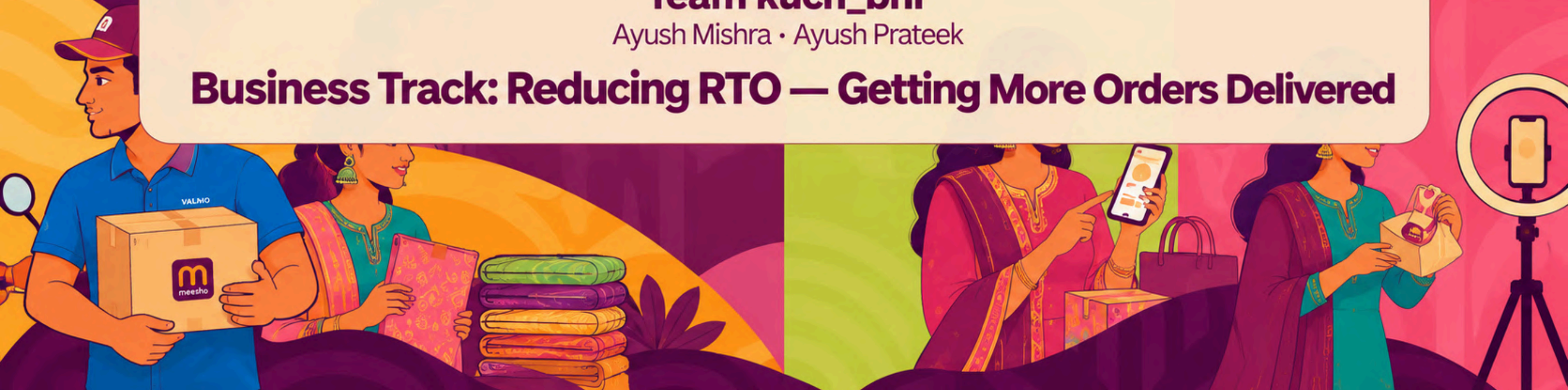




Team kuch_bhi

Ayush Mishra • Ayush Prateek

Business Track: Reducing RTO — Getting More Orders Delivered



Executive Summary: From Insight to Action

RTO is not one failure point. It builds across customer intent, address quality and last-mile execution.

01 THE PROBLEM COD concentrates RTO volume



94%
of illustrative RTO
volume is COD



16 of 17
RTOs per 100 orders

Source: Case data pack (illustrative).

What customers say they would choose after a missed delivery

46%

Reschedule

28%

Alternate receiver

18%

Wait for another
delivery

6%

Self-collect

Source: Survey, 67 COD users (73 respondents).

02 THE RESPONSE A two-part plan



RISK ENGINE

Spot risky COD orders
before dispatch.
Nudge once.
Never block.



RECOVERY PLAN

Give a failed delivery
a second chance
before the ₹120 return.

Our estimate: ~3.3 fewer RTOs per 100 orders (17 → ~13.7).
Range 1.7–5.0. A hypothesis to test in the pilot.

03 BUILT FROM EVIDENCE

WE ASSUMED



busy customers
would prefer to
collect parcels
themselves.

WE FOUND



only **6%** chose
self-collection.

WE CHANGED



lead with reschedule
(**46%**) and alternate
receiver (**28%**).
Self-collect is a fallback.

04 WHY THIS FITS VALMO



No one in the chain pays for the fix:
rider pay untouched, hubs paid
on collection (proposed).



Starts with signals Valmo already
holds: payment type, address,
order history.



Every confirmed address makes
the next order on it safer.

05 WHAT WE WILL TEST

1

PREVENT

Score risky COD orders
before dispatch.

2

RECOVER

Give failed orders another
path to delivery.

3

MEASURE

RTO vs control, rescue rate,
cost per rescue, customer
complaints.

06 THE QUESTION WE ANSWER NEXT

Where exactly does RTO happen, and **which failures should we solve first?**

CUSTOMER INTENT



ADDRESS



LAST MILE



DELIVERY OUTCOME

NEXT →

Diagnose the failure points
before choosing the fixes.

WHERE AND WHY RTO HAPPENS

RTO is not one failure point. It builds across the delivery journey.

01 CUSTOMER INTENT

Most orders are placed as COD.



94%
of illustrative RTO
volume is COD



16 of 17
RTOs per 100 orders

Source: Case data pack.

02 COD DECISION

80
COD
(80% of orders)

20
Prepaid
(20% of orders)

16
RTO (20%)

1
RTO (5%)

Source: Case data pack.

03 ADDRESS

Common, but rarely the final cause.



82% of COD users say a rider has had trouble finding or reaching them (66% say sometimes or more)

Source: Survey 1, 38 COD users.



Only **8%** name address as the main reason for their latest failed delivery (19% with 'couldn't be reached').

Source: Survey 2, 53 failed deliveries.



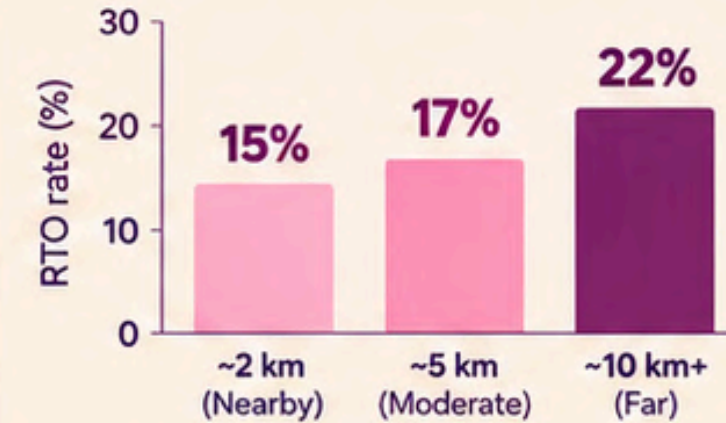
If the rider can't find the address, customers choose:

Share live location **40%**
Explain by phone **31%**

Source: Survey 2, 67 COD users.

04 LAST MILE

RTO rate by distance from hub.



Far orders fail about 1.5x as often as nearby ones.



₹21 of ₹50 (42%)
is last-mile delivery cost, the costliest leg.

Source: Case data pack.

05 DELIVERY OUTCOME

Every RTO pays twice.



₹170
per RTO
(₹50 forward + ₹120 reverse)



₹2,890
per 100 orders
(17 RTOs × ₹170)

Source: Case data pack.

06 WHY DELIVERIES FAIL

Reason for most recent failed COD delivery (n=53).



Source: Survey 2.

Not home + unreachable = 55% (29 of 53)

Why customers refuse (n=22, directional):
77% give behavioural reasons: changed mind 32%, cheaper elsewhere 23%, unsure of product 18%. Rounded.

Source: Survey 1.

07 EVIDENCE BACKBONE



CASE DATA

- 94% of RTOs are COD
- 80/20 COD-prepaid
- ₹21 of ₹50 last mile
- distance 15% → 22%



PRIMARY RESEARCH

- Survey 1: 50 respondents, 38 COD users.
- Survey 2: 73 respondents, 67 COD users.



APP TESTING

- Address auto-fill works with GPS, pincode, city, state.
- House No. and Building Name stay free text and unvalidated.



REFERENCE POINT

- Amazon Shipping India blog: evening slots associated with 47–53% lower RTO (vendor data, correlation).



Not yet sized: wrong-hub routing • rider intent • fake attempts. We log verified attempts (call log + geotag) and size them in the pilot.

08 KEY TAKEAWAY

COD creates exposure. Availability and reachability, 55% of failures, decide whether it becomes RTO.

NEXT →

Choose which failure points to solve first.

WHAT SHOULD WE SOLVE FIRST?

Prioritise where the evidence is strongest and the fix is practical.

01 HOW WE PRIORITISE

We prioritise interventions that create maximum impact without compromising key guardrails.



1 IMPACT

How much RTO exposure can this address?



2 FEASIBILITY

Can Valmo test it quickly with signals it already holds?



3 CUSTOMERS + OPERATIONS

Does it avoid adding friction for customers, riders or hubs?

02 P0 – PRE-DISPATCH RISK ENGINE

Acts before any delivery attempt.

PRIORITY: HIGH



WHY FIRST

Targets COD, where 94% of RTO volume sits.



EVIDENCE

94%

of RTO volume sits in COD.

Source: Case data pack

77%

of refusers give behavioural reasons (n=22, directional).

Source: Survey 1.



IMPACT

Measured in the pilot, not sized yet.

03 P1 – RECOVERY AFTER FAILURE

Gives a failed order a second chance.

PRIORITY: HIGH
Launches with P0.



EVIDENCE

46%

reschedule

28%

alternate receiver

Source: Survey 2, 67 COD users.

55%

of failures are not-home or unreachable (29 of 53).



IMPACT

16 COD RTOs x 55% not home or unreachable x 75% choose reschedule or alternate receiver x 50% haircut for stated vs actual bias will cut it.

~3.3 fewer RTOs per 100 orders (~19%).

Each rescue saves ₹120 – ₹21 = **₹99.**



CAVEAT

Survey asked about failed deliveries, not RTOs. Range 1.7–5.0 at a 25–75% haircut.

04 P2 – OTHER LEVERS, TEST LATER

Additional levers with potential, to be validated.

PRIORITY: VALIDATE



Prepaid incentive:

moving 25% of COD orders to prepaid ≈ 3 fewer RTOs per 100 (20 orders x 15-point gap). Naive; selection bias will cut it.



Hub-hold for self-collect:

only 6% chose it, so fallback only.



PRIORITY:

Build the risk engine:

Distance-from-hub signal,

once Valmo hub data is available.

05 KEY TAKEAWAY

Prevent early, recover the rest: **each rescued order turns a ₹120 return into a ₹21 re-attempt.**

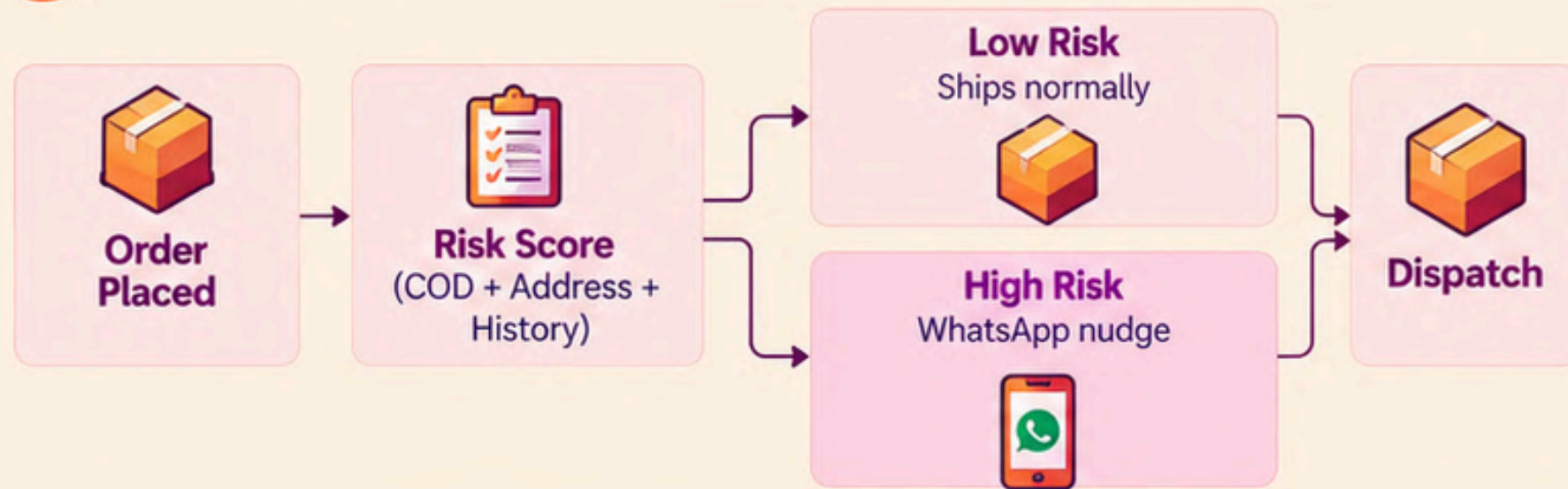
NEXT →

Build the risk engine.

Risk Engine: Prevent Failures Before Dispatch

A simple pre-dispatch engine that scores every order on 3 signals and sends risky orders one reason-specific WhatsApp nudge, without blocking, delaying or touching rider earnings.

01 HOW IT WORKS



Order Placed → Risk Score (COD + Address + History) → Low Risk: ships normally / High Risk: WhatsApp nudge → Dispatch

02 THE 3 SIGNALS

- 1 Payment type**
COD or prepaid.
- 2 Address check, 5 checks:**
 - pincode exists (official pincode directory)
 - pincode matches city/state
 - pincode is COD-serviceable
 - house no./building completeness
 - address succeeded before for this customer (bonus signal only: passes or stays neutral, never a failure).
- 3 Refusal history**
 - share of RTOs in the last 10 orders (design choice, calibrated in pilot).
 - First-time customers have no history.

03 RISK TIERS

Address Risk Tier

Failures	Tier
0	Clean
1 failure (excluding checks #1 or #3)	Medium
2+ failures, or check #1 or #3 fails alone	High (structural)

Customer History Tier

Refusal rate (last 10 orders)	Tier
No history	Reliable (default)
0–1 refusals	Reliable
2+ refusals	At-Risk

New customers are never flagged for having no history. This is our policy choice; the case pack notes new addresses fail more.

04 FINAL DECISION TABLE

#	Payment	Address	History	Action
1	Prepaid	Any	Any	Ships normally
2	COD	Clean	Reliable or New	Ships normally
3	COD	Medium	Reliable or New	WhatsApp: confirm house no. / landmark
4	COD	High (structural)	Reliable or New	WhatsApp: correct address before dispatch
5	COD	Clean	At-Risk	WhatsApp: confirm intent + offer reschedule
6	COD	Medium or High	At-Risk	One combined WhatsApp: confirm intent + correct address

05 WHAT IS IN THE FIRST VERSION

Signal	Evidence basis	Status in v1
COD status	94% of RTOs are COD (case pack)	Included
Address checks	Survey 1: 82% had rider trouble; app testing	Included
Refusal history	Buildable signal, needs Valmo calibration	Included
Distance from hub	15% → 22% in case pack, but no Valmo hub data yet	Later

06 GUARDRAILS

- 1 Never block or cancel an order.**
Every nudge is an offer.
- 2 Rider pay untouched.**
Everything happens before dispatch.
- 3 Cheap by design:** a WhatsApp message costs about ₹0.12 (utility) to ₹0.86 (marketing) plus GST, against ₹170 per RTO.
- 4 New customers are never flagged** only for having no history.
- 5 One message per order.**
If ignored, it ships as normal COD.

07 EXPECTED EFFECT

Impact is measured against a same-size control group in the pilot. Scoring weights are illustrative until calibrated on Valmo data.



Our Working Prototype

08 KEY TAKEAWAY

Don't predict perfectly. **Detect early, explain the risk, nudge once, and never block.**

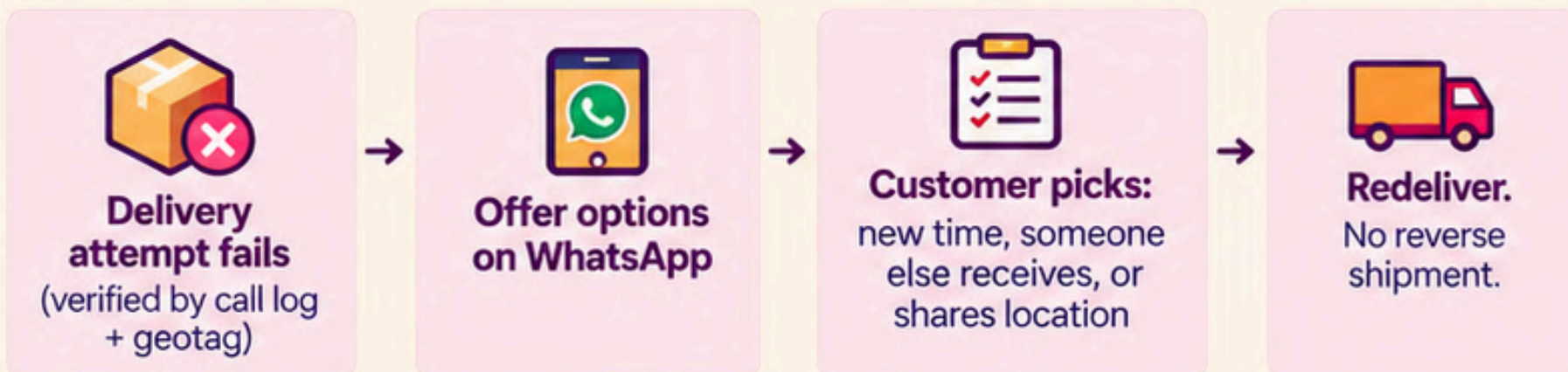
NEXT →

Recovery plan: failures that still happen.

Refused-Order Recovery: A Second Chance Before the ₹120 Return

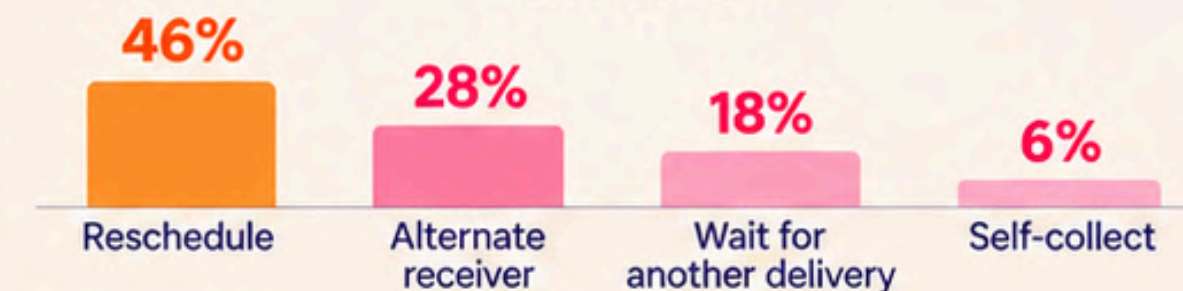
Prevention won't stop every failure. Recovery gives a refused or undelivered order a second chance inside a fixed window, before it becomes a reverse shipment.

01 WHEN RECOVERY STARTS



02 WHAT CUSTOMERS TOLD US

What customers prefer after a missed delivery (our research)



75% chose reschedule or alternate receiver.
Only **6%** chose self-collect, so it is a fallback, not the headline.

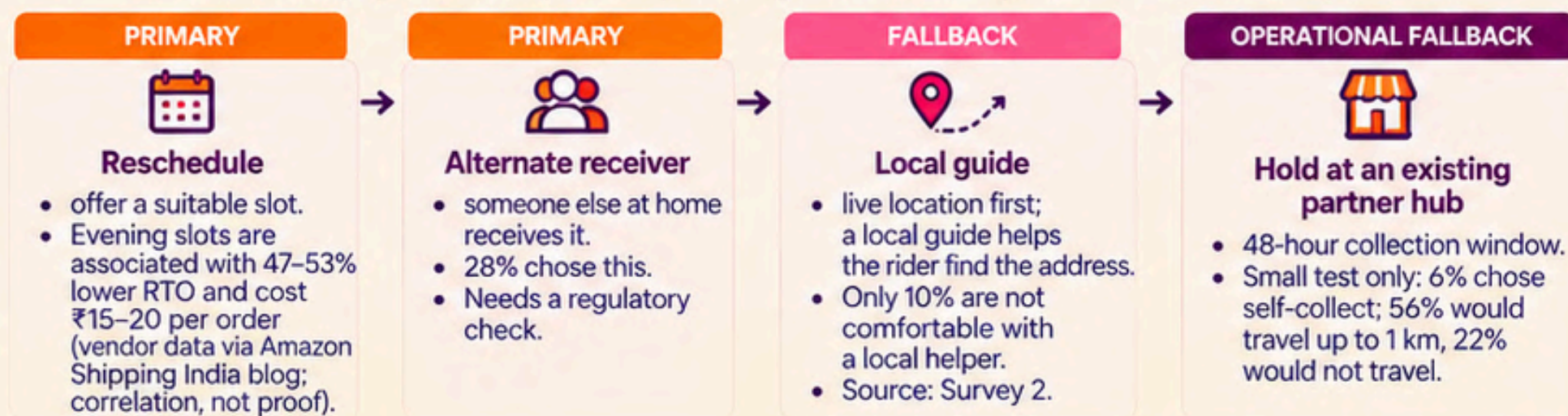
Source: Survey 2, 67 COD users.

If the rider can't find the address:



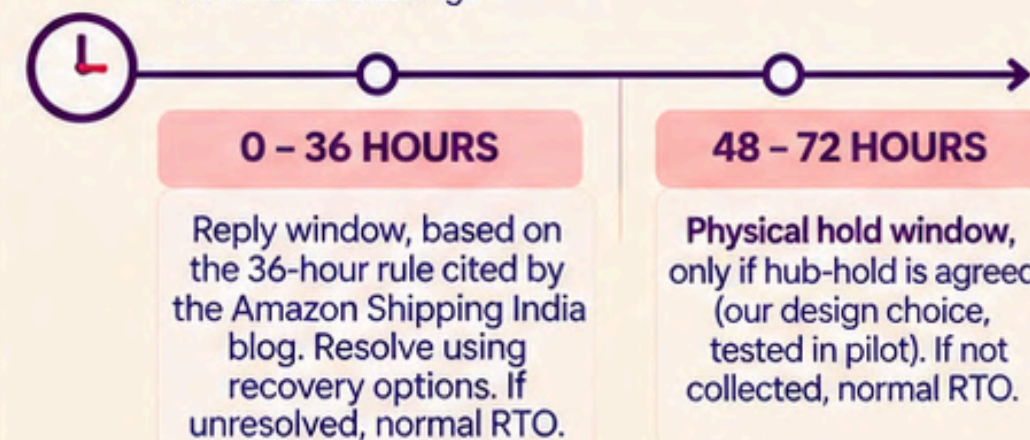
03 THE RECOVERY LADDER

We offer recovery options in a sequence — prioritising what customers prefer and what can be implemented.



04 THE RECOVERY CLOCK

Recovery happens within a defined window — no endless chasing.



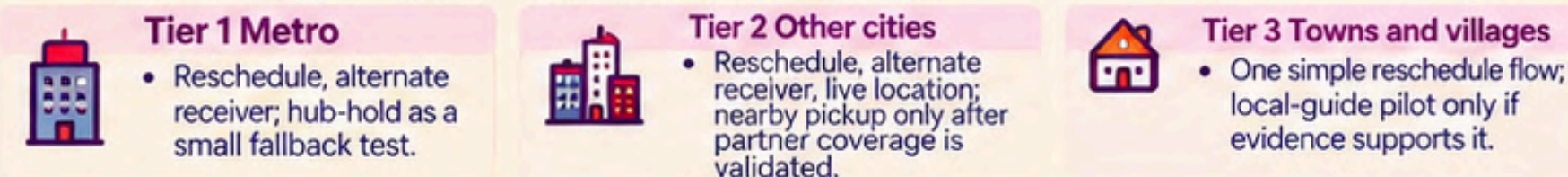
05 COST CEILING



Source: Case data pack.

06 TIERED LOGIC

Different geographies need slightly different recovery options.



07 IF RECOVERY FAILS

- Window ends → normal RTO.
- No repeated messages.
- No endless chasing.

08 HOW WE MEASURE

- Offer acceptance by option
- Redelivery success
- Cost per rescued order
- Customer complaints
- Rider impact
- RTO vs control.

09 KEY TAKEAWAY

Don't let one failed attempt become an automatic ₹120 return. Offer a ₹21 second chance first.

NEXT →
User cohorts: one system, different defaults.

User Cohorts: Different People, Different Defaults

Survey 2: 73 respondents, 67 COD users (6 who never use COD excluded). Four behaviour groups, one recovery system.

01 METRO HABIT (n=27)

Reschedule is the strongest preference.



48%

Reschedule



30%

Alternate receiver

Address help:



33%

live location



33%

phone call

Default:

Reschedule + alternate receiver.

02 OTHER-CITY HABIT (n=15)

Alternate receiver is more common.



40%

Alternate receiver



33%

Reschedule



60%

prefer live-location sharing.

Primary address fix:

Live location.

03 TOWN / VILLAGE HABIT (n=8)

Rescheduling dominates.



75%

reschedule
(6 of 8).

Directional, small sample.

Default:

One good reschedule flow.

04 NEW / OCCASIONAL (n=17)

Needs more support, not less.



41%

Reschedule



24%

Alternate receiver



35%

would not travel to collect (6 of 17).



Local guide:
3 of 17, directional.



Cancel:
1 of 17, on the missed-delivery question.

No history is not a risk signal.
This is our policy choice.

05 FROM RISK TO ACTION

Different failure reasons need different recovery options.

Failure reason		What it means	→	Recovery action
Address issue	→	Incomplete or unclear address	→	Help fix the address
Not home	→	Customer wasn't available	→	Reschedule
Can't receive	→	Prefers someone else to receive	→	Alternate receiver
No history	→	New or occasional customer	→	No penalty

06 HOW WE USE THIS

One recovery system, with cohort-specific defaults.



Cohorts set the default offer, not a separate product.



Every option is built from what customers told us.



Small groups (n=8, n=17) are tested in the pilot before we rely on them.

07 KEY TAKEAWAY

One recovery system, different defaults, chosen by **how customers actually behave.**

NEXT →

30-60-90 day test plan.

30-60-90 DAY TEST PLAN

Test small. Measure against a control. Scale only what clears the gate.

01 DAYS 1-30

PROVE



Risk Engine + reschedule and alternate-receiver nudges live in one pilot region, against a same-size control group.

LAUNCH



Risk Engine (COD, address checks, history)



Reschedule and alternate-receiver options on WhatsApp



Verified-attempt logging (call log + geotag)

MEASURE



Risk-scoring coverage



Nudge response rate



Pilot RTO vs control



Customer complaints



Hypothesis to test:

~3.3 fewer RTOs per 100 orders (range 1.7-5.0). A hypothesis, not a claim.



Owner:

Valmo ops lead.



Metric:

RTO rate vs control.



GATE → PHASE 2

Stable system + no rise in complaints + directional RTO improvement vs control. If not, stop and fix.

02 DAYS 31-60

EXPAND



Expand to 2-3 regions. Validate Tier 2. Test hub-hold as a small fallback at existing partner hubs.

₹120

REVERSE SHIPMENT COST

—

₹21

EXTRA LAST-MILE ATTEMPT

=

₹99

RECOVERY COST CEILING



Test pickup or digital-payment recovery only after partner coverage is validated.

MEASURE



RTO vs control



Rescue rate



Cost per rescued order



Gaming signals



Owner:

Valmo finance + hub partners.



Metric:

cost per rescue under ₹99.



GATE → PHASE 3

Cost per rescue under ₹99 + no operational or fairness issue.

03 DAYS 61-90

DECIDE



Local-guide pilot in towns only if survey comfort and earlier results support it. Full evaluation across all pilots.

Prevention



Risk Engine

+

Recovery



Tier 1/2/3

VS.

Same-size control



EVALUATE ACROSS PILOTS



RTO reduction vs control



Cost per rescue



Customer complaints



No gaming or abuse

FINAL DECISION (FOR EACH RECOVERY TIER)

SCALE



Evidence clears the gate.

ITERATE



Promising, needs refinement.

HOLD



Does not justify expansion.



Owner:

Meesho + Valmo product lead.



Metric:

net RTOs per 100 orders vs control.

04 KEY TAKEAWAY

Every expansion is gated by **evidence, not assumption.**

PILOT



MEASURE



GATE



EXPAND

NEXT →

Risks, guardrails and open questions.

Risks, Guardrails & Open Questions

A good solution doesn't hide uncertainty. It puts guardrails around it.

01 KEY RISKS AND FIXES

Risk	Why it matters	Mitigation
 Riders use hold-at-hub as a shortcut	Hold could replace real delivery attempts	Unlock hold only after logged call attempts + geotag.
 Fake or weak delivery attempts	Create false RTOs and unhappy customers	Log as a named reason code. Hub partner accountable through call log + geotag audit.
 Customers abuse reschedule	Endless reschedules delay delivery	One reschedule and one alternate-receiver change per order.
 Alternate-receiver delivery may have regulatory limits	Recovery path may not be production-ready	Open validation item; confirm with Meesho before scaling.

02 NON-NEGOTIABLES

COST	RIDERS	CUSTOMERS
 Recovery must cost less than the return it replaces. ₹99 ceiling.	 Rider pay untouched. Prevention happens before dispatch. Hubs paid on collection (proposed).	 No checkout friction. Scoring happens after order placement. New customers are not penalised for having no history.

03 TO CONFIRM WITH MEESHO

-  Does Valmo already run NDR outreach or re-attempts?
-  Can Valmo hubs physically hold parcels for collection?
-  Does the "Latent Last Mile" program exist, and where?
-  What is PayNearby / Spice Money's actual Tier 2 pincode coverage?

04 WHAT THE DATA DOES NOT SUPPORT

 No Valmo hub-level data, so no hub-performance risk layer.	 Address is common (82% have had trouble) but rarely the final cause (8%), so it is not our headline.	 Prepaid impact is a naive estimate; selection bias not sized.	 Town/Village (n=8) and local-guide preference (3 of 17) are directional only.	 Wrong-hub routing, rider intent and fake attempts are not sized; we measure them in the pilot.	 No national result yet, so no national RTO reduction claimed.	 Dropped from Round 1: the assumption that busy customers prefer self-collection (only 6% chose it).
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05 KEY TAKEAWAY

We name what we don't know, then **design the test to find out.**

RISK



GUARDRAIL



VALIDATION



SCALE

NEXT →

Sources used.

Sources Used in the Deck

A mix of case data, primary research and external references.

01 Given / Primary (Meesho-provided)



Meesho DICE Challenge S3 Case Brief + Data Pack

COD/prepaid RTO rates, forward/reverse logistics cost, distance-vs-failure table.

02 Our Own Primary Research



Survey A — “Online Shopping & COD”

n=50 unique, 38 COD users
Refusal reasons, address difficulty, certainty at ordering.



Survey B — “Customer Delivery Experience”

n=73 unique, 67 COD users
Recovery preferences, cohort behaviour, address-finding preference.



Direct Meesho app testing

GPS-on/GPS-off address field validation gap.



Driver/delivery-partner interviews

Qualitative failure-point ranking.

03 Secondary Data Analysis



Kaggle: meesho_supplier_returns_data.csv

789 orders, single seller

16-day median reverse-cycle time.

04 Meesho’s Own Financial & Operational Disclosures



Screener.in, sourcing Meesho’s IPO/RHP prospectus

Valmo in-sourcing share (~2%→64%), COD share trend, NMV/GMV.



Financial Express / Inc42 / legalparivar.com

Valmo order share (~62%), fulfilment cost (₹50.45→₹37.70), 19,000 pin codes.



Valmo/Meesho job postings (Lever.co, startup.jobs)

Meesho Pickup Point (prepaid-only), “Latent Last Mile” program.



Meesho’s own “11 Mantras” post and “Listen or Die” company blog/CPO interview

Culture framework behind the cohort methodology.

05 Competitor & Industry Benchmarking



Shiprocket — official RTO Score support doc, RADAR blog, Engage platform.



Delhivery — RTO Predictor official product page.



Amazon India — “I Have Space,” Society Pickup Points, EMNLP 2022 address-ML research paper.



Flipkart — eKart kirana network, IVR confirmation (iThinklogistics), WWW’23 academic paper on RTO detection.



Myntra — Mensa Network (Entrackr).



Amazon Shipping India NDR blog — 36-hour NDR window, evening-slot RTO reduction, fake-attempt reason code.



GoKwik, Qikink, Razorpay — RTO industry blogs (treated as directional/vendor-sourced).



Jumia, Mercado Libre, Cainiao — international pickup-point precedents.



OXO Pay (Mexico), Indomaret/Alfamart (Indonesia) — cash-to-digital pay-at-store precedents.



PayNearby / Spice Money — cash-collection agent network claims (flagged unverified, pending mentor confirmation).



India Post Pincode API — free government address-validation source.

06 Your Own Prior Work



“Latent Personas” deck (your past DICE team)

Fears → Design framework, Kirana Connect model, adapted for Tier 3 recovery.

07 Explicitly Excluded (for transparency)



Indian_RTO.csv

(wrong dataset — vehicle registration, not logistics RTO)



Daraz, InPost/Poland

(ruled out — insufficient public detail / wrong market structure)



Unverified hub/sorting-center/order-volume figures from the Latent Personas deck

(dropped, couldn’t independently confirm).



Thank You!

Thank you for your time and consideration.

We look forward to your feedback and the opportunity to discuss our approach.